

Hybrid Music Clustering with Multi-Modal VAEs on GTZAN

CSE425 VAE Project Report

January 7, 2026

Abstract

We study latent representation learning for hybrid music clustering on the GTZAN dataset by integrating audio features and lyrics. We implement and compare a basic Variational Autoencoder (VAE), a Hybrid VAE (audio+lyrics), a Conditional VAE (CVAE), and Beta-VAE variants, and evaluate clustering quality with Silhouette, Calinski–Harabasz, Davies–Bouldin, Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and Purity. Across experiments, Hybrid VAE and VAE latents improve clustering over PCA baselines; Beta-VAE with moderate β offers some disentanglement but mixed clustering gains, while CVAE shows modest improvements given limited data.

1 Introduction

Music genre clustering benefits from compact, semantically meaningful embeddings derived from audio and lyric modalities. Variational Autoencoders (VAEs) provide a principled framework to learn such latent spaces by maximizing the evidence lower bound (ELBO). We explore non-linear latent spaces and modality fusion to improve genre clustering quality compared to linear baselines.

2 Related Work

VAE (Kingma & Welling, 2014) formalizes deep latent-variable models trained with ELBO. Beta-VAE (Higgins et al., 2017) scales the KL term to encourage disentanglement. CVAE (Sohn et al., 2015) conditions the generative process on auxiliary information. For music representation, learned embeddings from audio (e.g., MFCCs, spectral features) and lyrics are widely used; clustering methods include K-Means, Agglomerative, and density-based approaches.

3 Method

3.1 VAE Architecture

Implemented in `src/models/vae.py`, our models use MLP encoders/decoders for tabular features. BasicVAE encodes audio features into a latent z (e.g., hidden dims [512, 256, 128], latent dimension 128). HybridVAE employs dual encoders (audio and lyrics) to a shared latent and dual decoders for reconstruction. CVAE concatenates inputs with a one-hot genre condition. Beta-VAE scales the KL divergence by $\beta > 1$.

3.2 Objective

We optimize the ELBO:

$$\mathcal{L}(x; \theta, \phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \beta \text{KL}(q_\phi(z|x) \| p(z)), \quad (1)$$

where $\beta = 1$ recovers the standard VAE; $\beta > 1$ yields Beta-VAE.

3.3 Feature Extraction (Audio + Lyrics)

Audio features (e.g., MFCCs, spectral rolloff, chroma) and lyric-derived features are combined or conditioned depending on the variant. Dataset details and EDA are in `notebooks/eda.ipynb`; raw data is under `data/GTZAN`.

3.4 Clustering Methods

We cluster latent vectors using K-Means and Agglomerative Clustering; in the medium track, DBSCAN is also considered. Baselines include PCA+K-Means and Autoencoder+K-Means.

4 Experiments

Dataset: GTZAN with 10 genres: blues, classical, country, disco, hiphop, jazz, metal, pop, reggae, rock. Easy track uses audio features for ~ 609 samples; hybrid tasks use audio+lyrics for ~ 122 samples.

Preprocessing: Standardization of features; train/eval splits where applicable; missing value checks and file existence verification.

Training: `easy/main.py`, `medium/main.py`, `hard/main.py` configure epochs (e.g., 50–150), learning rates (5×10^{-4} to 10^{-3}), and batch sizes (e.g., 16). Latents are clustered post-training.

5 Results

5.1 Easy: BasicVAE vs. PCA Baseline

Metrics from `easy/results/metrics_comparison.csv`.

5.2 Medium: HybridVAE + Multiple Clusterers

Metrics from `medium/results/clustering_comparison.csv`.

5.3 Hard: CVAE, Beta-VAE, and Baselines

Metrics from `hard/results/comprehensive_comparison.csv(selectedmethods)`.

6 Discussion

VAE latents improve clustering metrics over PCA, indicating non-linear embeddings capture genre-relevant structure from audio features. HybridVAE leverages lyrics to further increase Silhouette and NMI under Agglomerative/K-Means. In the hard setting, Beta-VAE with $\beta = 4.0$ shows moderate latent separation but does not surpass hybrid clustering;

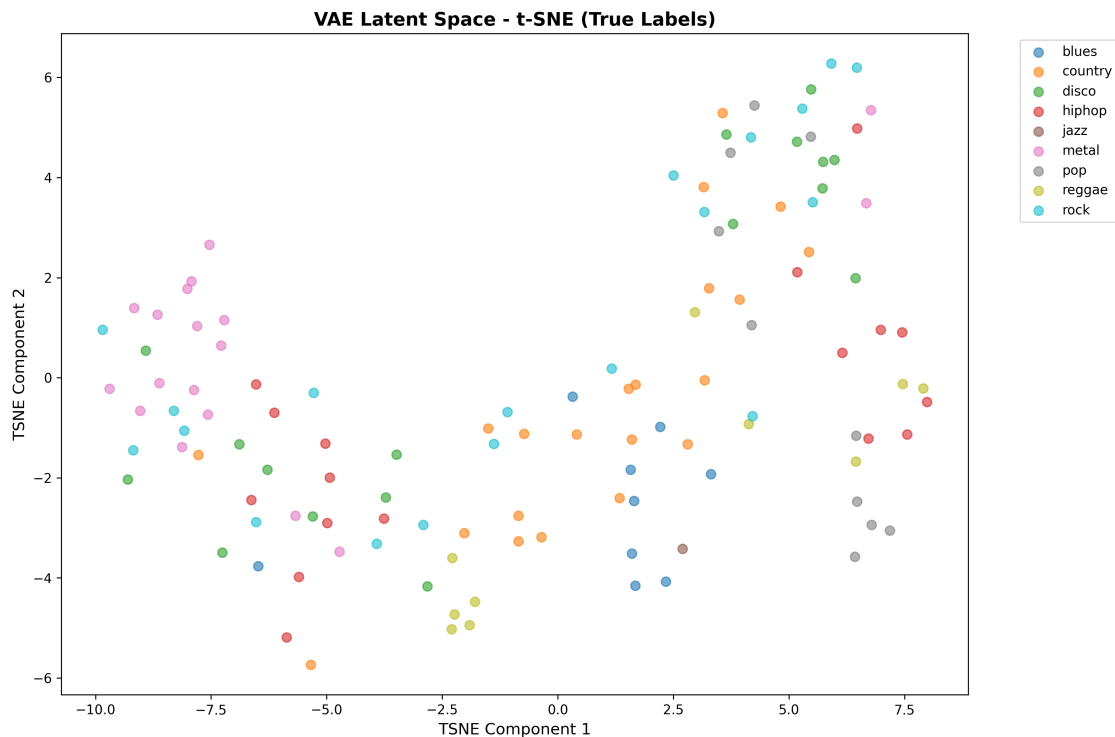


Figure 1: VAE latent space t-SNE with true labels (Easy).

CVAE conditioning yields modest gains, likely limited by sample size (~ 122) and modality mismatch.

7 Conclusion

Multi-modal VAEs provide promising improvements for genre clustering on GTZAN. HybridVAE offers the best overall balance in this project. Future work includes larger datasets, spectrogram-based ConvVAEs, improved lyric embeddings, and end-to-end multimodal training.

References

References

- [1] Diederik P. Kingma and Max Welling. Auto-Encoding Variational Bayes. *ICLR*, 2014.
- [2] Irina Higgins et al. beta-VAE: Learning Basic Visual Concepts with a Constrained Variational Framework. *ICLR*, 2017.
- [3] Kihyuk Sohn, Honglak Lee, Xinchen Yan. Learning Structured Output Representation using Deep Conditional Generative Models. *NIPS*, 2015.
- [4] George Tzanetakis and Perry Cook. Musical genre classification of audio signals. *IEEE TASLP*, 2002.

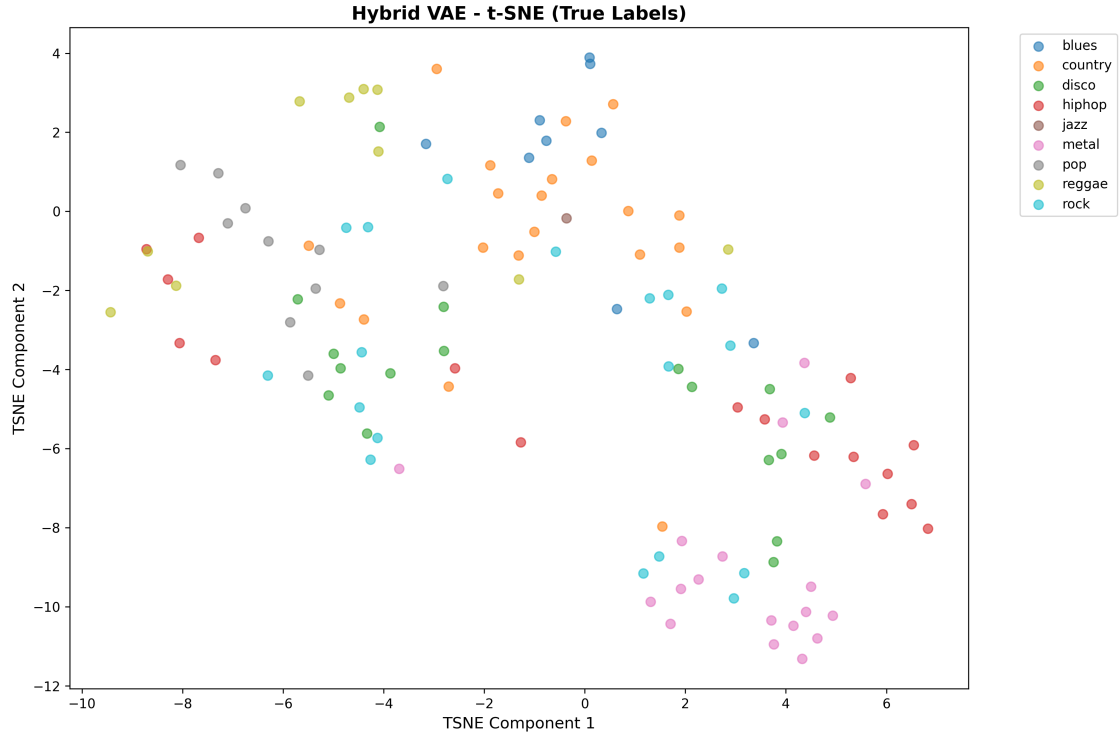


Figure 2: HybridVAE latent space t-SNE (Medium).

[5] Stuart Lloyd. Least squares quantization in PCM. *IEEE*, 1982.

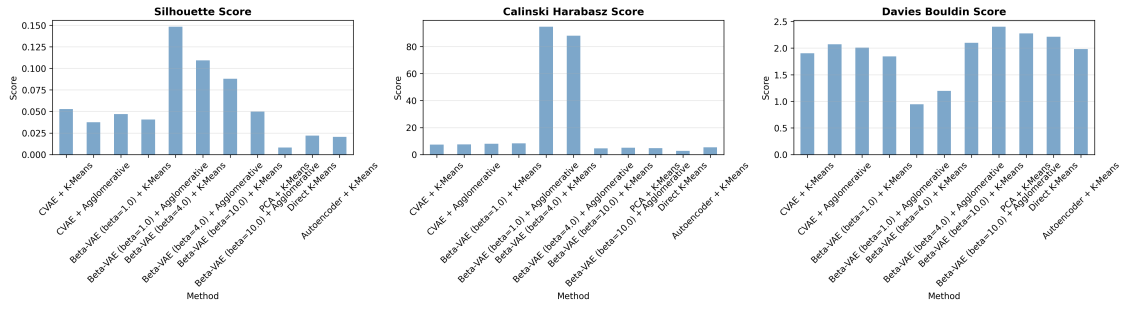


Figure 3: Comprehensive comparison across methods (Hard).